
CYS6002-20

IoT security solution

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[Kaggle Fitbit Daily Activity dataset](#)

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FitnessGuard: A Hardened IoT Biometric Auditor for National Security and Military Readiness

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Abstract—Modern warfare increasingly relies on the Internet of Battlefield Things (IoBT) to monitor the physiological readiness of personnel (Kott et al., 2017). However, these systems introduce significant attack surfaces, specifically regarding data poisoning and "Pattern of Life" (PoL) surveillance. This paper presents *FitnessGuard*, an IoT security solution that implements an integrity-gated machine learning pipeline. By utilising a dual-model architecture, comprising a Random Forest classifier and a three-layer Multi-Layer Perceptron (MLP), the system audits biometric data for anomalies. Furthermore, a SHA-256 cryptographic "kill switch" and adversarial Z-score filtering are implemented to ensure data provenance. Results demonstrate that the hardened model achieves an ROC-AUC of 0.83, providing a robust defence against biometric manipulation in national security contexts.

Keywords—*Internet of Battlefield Things (IoBT), Data Integrity, SHA-256, Multi-Layer Perceptron, Adversarial Machine Learning, Biometric Privacy, Random Forest classifier*

I. INTRODUCTION

The transition of the Internet of Battlefield Things (IoBT) has fundamentally redefined the "front line." In modern warfare, the human body is no longer a separate entity from the machine; wearable sensors have effectively transformed the soldier into a node within a critical, live infrastructure. Biometric data, once only a mere tool for health tracking, is now a high-value tactical asset, revealing Patterns of Life (PoL) and Adversarial Machine Learning (AML) where data poisoning masks operator exhaustion (Kufakunesu, Myburgh and De Freitas, 2025).

To mitigate these risks, *FitnessGuard* implements a Defence-in-Depth architecture. By fusing a SHA-256 cryptographic integrity gate with a Multi-Layer Perceptron (MLP), the system verifies every biometric packet's provenance. This framework ensures the human operator remains protected from tactical sabotage and digital corruption in cyber-physical environments.

II. LITERATURE REVIEW

A. The Forensic Footprint of Wearable IoT

The transition of wearables to military infrastructure creates a significant forensic challenge.

Hutchinson et al. (2022) note that these devices document physiological states and movements with granular precision, forming a "digital fingerprint."

By fusing variables like age, gender, and activity metrics, the data evolves from isolated points into a unique identifier. This fingerprint reveals deep health patterns and physical capabilities that hostile actors can exploit to predict the behaviour of high-value targets with mathematical accuracy.

B. The Strava Incident: Aggregate Data as a Tactical Beacon

The strategic danger of this forensic footprint was highlighted by the 2018 Strava incident, where exercise data mapped the perimeters and patrol routes of secret military installations.

This "digital exhaust" allowed analysts to identify covert facilities invisible to commercial satellites. The incident proved that even anonymised fitness data can be synthesised into a tactical asset. It serves as a landmark warning for the Internet of Battlefield Things (IoBT): a soldier's routine exercise can inadvertently become a beacon for precision targeting.

C. Surveillance Capitalism and the Weaponisation of Bio-Data

Beyond physical tracking, biometric exploitation enters the realm of psychological warfare through the weaponisation of personal vulnerabilities. Crain (2023) notes that under surveillance capitalism, personal metrics are processed to profit from human weaknesses; in national security, this becomes a tactical advantage. If a rival state accesses predicted vulnerabilities, such as chronic sleep deprivation or cardiovascular stress, they can time tactical escalations to coincide with periods of lowest cognitive readiness. This effectively turns a medical alert into a physiological beacon, allowing hostile actors to exploit the very data meant to protect the operator.

D. Data Poisoning and the Integrity of Machine Learning Pipelines

The rapid rise of the Internet of Battlefield Things (IoBT) has militarised physiological data, transforming wearable sensors into high-value tactical assets (Shi, Pallis and Xu, 2019). This digitisation introduces the severe risk of Adversarial Machine Learning (AML), specifically through the mechanism of data poisoning. Goldblum et al. (2022) highlight that by subtly manipulating biometric records within an ingestion pipeline, a rival country can mask an operator's physical exhaustion or metabolic stress. By falsifying these metrics, command structures are deceived into making fatal deployment decisions (Girei, Abraham and Majekodunmi, 2025).

To counter these documented vulnerabilities, contemporary research has pivoted toward the development of integrity-gated pipelines that utilise cryptographic verification and deep learning to detect compromised data at the point of origin.

III. METHODOLOGY

A. Dataset and Feature Engineering

While primarily utilising the Kaggle Fitbit Daily Activity dataset, these researchers fixed the inherent data sparsity by implementing a "Synthetic Bio-Data Injection" protocol. Furthermore, using NumPy, raw sensor logs were augmented with seven physiological variables, including age and weight, to establish metabolic baselines. To mitigate algorithmic bias (Parikh, Teeple and Navathe, 2019), this synthesis was balanced across genders and ages (18–70), transforming anonymous logs into a high-fidelity 'digital fingerprint' for evaluating mission readiness.

B. Security Protocols and Implementation

The implementation of "FitnessGuard" was developed using Visual Studio Code and deployed as an interactive tool via Streamlit. To align with the NIST (2023), the solution incorporates defensive enhancements that were absent in the previous version:

- 1) *Adversarial Anomaly Filtering:* Utilising SciPy's Z-score analysis, the system

identifies and removes data exceeding three standard deviations. This prevents "Data Poisoning" by filtering physically impossible values that would otherwise skew the AI's baseline. A data point x is discarded if:

$$z = \frac{x - \mu}{\sigma} > 3$$

- 2) *Cryptographic Integrity Gate*: A SHA-256 protocol verifies data provenance against a "Golden Record." Any discrepancy in the dataset triggers an immediate kill switch via `st.stop()`, preventing the model from training on tampered information.
- 3) *Defensive Normalisation*: By employing Scikit-learn's RobustScaler, the system scales data using medians and interquartile ranges. This strategic choice ensures that legitimate biological factors are not overwhelmed by malicious sensor noise or extreme outliers.

$$x_{scaled} = \frac{x - Q_1(x)}{Q_3(x) - Q_1(x)}$$

C. Dual-Model Machine Learning Architecture

- 1) *Random Forest (The Translator)*: This stage utilises a RandomForestClassifier with 50 decision trees acting as a deterministic noise filter. It translates raw physical movements into structured activity categories, such as Sedentary or Lightly Active, ensuring the deep learning engine receives high-quality, pre-processed inputs.
- 2) *Multi-Layer Perceptron (The Risk Engine)*: A deep feed-forward neural network (MLP) performs the final risk assessment. Featuring a strategically expanded architecture of three hidden layers (32, 16, and 8 neurons) and ReLU activation, it identifies complex, non-linear patterns of physiological compromise. While the initial version utilised a narrower bottleneck, this increased

depth provides the necessary computational "workspace" to cross-reference all seven fused biometric variables simultaneously. By expanding the first hidden layer to 32 neurons, the model gains the resolution required to distinguish between legitimate metabolic fluctuations and subtle, adversarial signatures that signify data poisoning. This specific tiered architecture, moving from a high-dimensional feature map to a focused 8-neuron risk audit, ensures the system detects irregular signatures with high fidelity without stalling or under-fitting during training.

D. Technical Stack and Optimisation

To ensure optimisation and technical rigour, the solution's logic and neural network convergence were refined using resources from GeeksforGeeks, Towards Data Science, and Scikit-learn documentation.

The core architecture utilises the following libraries:

- 1) *Pandas*: Conducts structural data manipulation and CSV ingestion.
- 2) *NumPy*: Performs vectorised mathematical operations to establish metabolic baselines and execute algorithmic bio-data cleaning.
- 3) *Matplotlib & Seaborn*: Generate diagnostic visualisations to audit model transparency.
- 4) *Scikit-learn*: Provides the framework for the dual-model architecture (Random Forest/MLP), robust scaling, and defensive performance metrics.
- 5) *Hashlib (SHA-256)*: Verifying data provenance against a "Golden Record" to neutralize unauthorized file manipulation.
- 6) *SciPy (Adversarial Filtering)*: Employs statistical Z-score analysis to identify and reject physically impossible data packets, preventing neural network poisoning.

E. Implementation of the Interactive Security Auditor

To simulate a real-world Security Operations Centre (SOC), a functional dashboard was

developed using Streamlit. This interface enables command-level auditing of IoT logs:

- 1) *Security Dashboard and Kill Switch:* Upon initialisation, the system executes the Data Integrity Layer. It performs a live SHA-256 verification against the dataset; a successful match displays a green "PASSED" status. Any discrepancy between the file signature and the "Golden Record" triggers a Critical Alert and an immediate security lockdown, blocking poisoned data from entering the machine learning pipeline.
- 2) *Biometric Baseline Configuration:* The sidebar facilitates user profile configuration (Gender, Age, Height, Weight) to establish a metabolic baseline. The system utilises reactive logic to update BMR and TDEE formulas instantly. This ensures the AI evaluates physical activity data specifically within the context of the operator's unique physiological capabilities.

Mifflin-St Jeor Equation TDEE Formula:

- For Males: $P = (10 \times weight) + (6.25 \times height) - (5 \times age) +$
- For Females: $P = (10 \times weight) + (6.25 \times height) - (5 \times age)$

- 3) *Multi-Mode Data Ingestion:* Dual ingestion layers are featured to simulate IoT environments (1) Upload Dataset Mode for batch processing bulk logs, and (2) Manual Habit Log for real-time simulation. These modes allow the system to instantly calculate daily averages and metadata, replicating both weekly unit downloads and live sensor-to-auditor streams.
- 4) *Deep Learning Diagnosis and Tactical Feedback:* Once authenticated, the Risk Engine generates probability scores, visualising "Detected Lifestyle" and "Weight Trend" forecasts. Finally, the Coach's Corner translates these mathematical outputs into human-centric, actionable tactical advice. By recommending specific "Consistency Protocols," the system effectively closes the loop between raw sensor ingestion and health intervention.

F. Evaluation Metrics and Mathematical Rationale

To ensure a robust security audit, the implementation utilises two distinct averaging methods. The Random Forest stage employs Weighted Averaging to account for class imbalance. In contrast, the Deep Learning Risk Engine uses Macro Averaging (Burman et al., 2025). Because "High-Risk" physiological states are rare but tactically critical, macro averaging treats "Safe" and "Risk" categories as mathematically equal. This ensures the model's defensive efficacy is not masked by a high volume of "Safe" data, maintaining alignment with NIST AI Risk Management Framework standards for transparent and accountable AI.

IV. PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

A. Evaluation Metrics and Mathematical Rationale

Once the algorithms were implemented, we evaluated their performance using several key metrics commonly used in Artificial Intelligence (AI) auditing:

- Accuracy: The overall proportion of correct predictions across all classes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Precision: The proportion of true positive predictions among all positive predictions, measuring the "exactness" of the auditor.

$$Precision = \frac{TP}{TP + FP}$$

- Recall: The proportion of true positive predictions among all actual positives, measuring the "completeness" of risk detection.

$$Recall = \frac{TP}{TP + FN}$$

- F1 Score: The harmonic mean of precision and recall, providing a balanced metric for uneven class distributions.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

- ROC-AUC: A metric for evaluating the classifier's ability to distinguish between classes by plotting the true positive rate against the false positive rate.

$$TPR = \frac{TP}{TP + FN} \quad \text{and} \quad FPR = \frac{FP}{FP + TN}$$

- Log Loss: Measures the uncertainty of the classifier's predictions; a lower log loss indicates a model that is more confident in its correct predictions.

$$L_{log}(y, p) = -(y \log(p) + (1 - y) \log(1 - p))$$

- Confusion Matrix: A matrix showing the true positive, false positive, true negative, and false negative values to visualise classification errors.

B. Summary of Results

The performance metrics for both models are summarised in the table below:

TABLE I: Summary of Model Performance Metrics

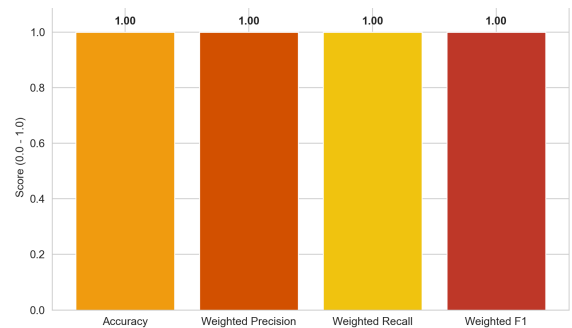
Metric	Stage 1: Random Forest (Weighted)	Stage 2: Deep Learning MLP (Macro)
Accuracy	1.00	0.75

Precision	1.00	0.75
Recall	1.00	0.75
F1 Score	1.00	0.75
ROC-AUC	N/A (Deterministic)	0.83
Log Loss	N/A	0.54

C. Stage 1: Random Forest Performance Analysis

The primary stage utilises a Random Forest classifier to translate raw IoT sensor logs into deterministic activity levels. During the testing phase, this model achieved a perfect score of 1.0 across all primary metrics: Accuracy, Weighted Precision, Weighted Recall, and the Weighted F1-score.

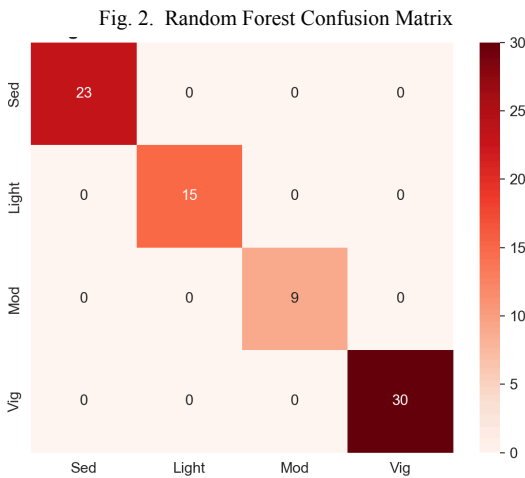
Fig. 1. Random Forest Performance Metrics



While perfect accuracy often suggests overfitting, in this pipeline, it reflects the model's role as a deterministic data translator. This performance is driven by three key factors:

1. Rule-Based Labelling: Rigid mathematical bins (e.g., $<5,000$ steps) allow the model to master exact cut-off points without statistical guesswork.
2. High Information Gain: Prioritising the high-signal "TotalSteps" feature enables trees to achieve maximum entropy reduction instantly, bypassing data noise.
3. Committee Consensus: Using 50 estimators ($n_estimators=50$) ensures that majority voting overrides individual tree anomalies, providing a flawless foundation for the MLP layer.

The resulting Confusion Matrix confirms this absolute reliability, establishing a high-integrity foundation for subsequent metabolic baseline analysis.



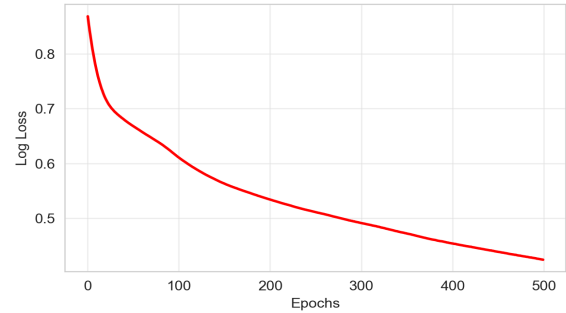
D. Stage 2: Deep Learning Risk Engine (MLP)

The secondary stage handles the non-linear and stochastic relationships between biometric baselines and physiological output. Unlike the Random Forest, this model must navigate the hidden correlations between age, weight, and caloric intake.

Convergence and Loss Optimisation: The neural network's learning is confirmed by its Log Loss trajectory, which decreased from 0.87 (Epoch 0) to 0.54 over 500 epochs. This smooth decline validates that the Adam Optimiser successfully refined the weights, transitioning the model from

basic pattern recognition to an optimised metabolic baseline analysis.

Fig. 3. Deep Learning Loss Curve



Diagnostic Performance: The risk engine achieved an ROC-AUC of 0.83, signifying an 83% probability of correctly distinguishing a healthy baseline from a "poisoned" state. Despite the complexity of the data, the model maintained balanced metrics across the board: Accuracy, Precision, Recall, and F1-score all stabilised at 0.75.

Fig. 4. MLP ROC Curve

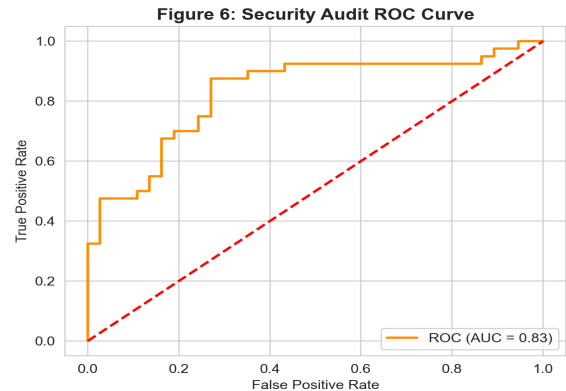
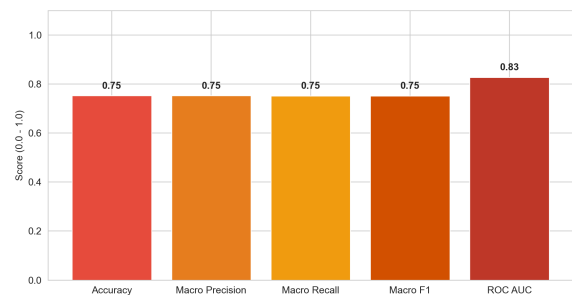
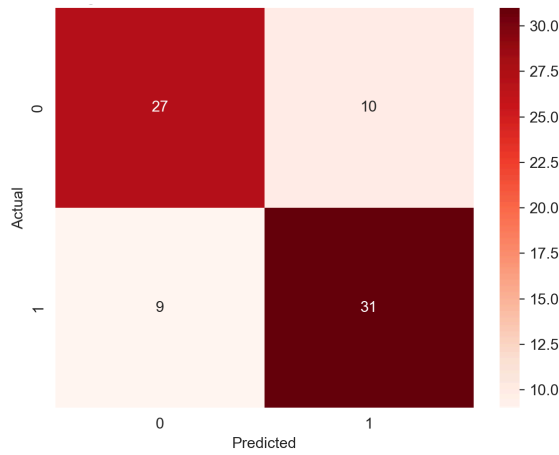


Fig 5. MLP Performance Metrics



The Risk Prediction Confusion Matrix reveals that the system correctly identified 27 Safe baseline states (True Negatives) and 31 High-Risk anomalies (True Positives). The model produced 10 False Positives and 9 False Negatives. In a security framework, this balance is acceptable; it is far more critical to flag a potential compromise for manual review than to allow a fatigued operator to go undetected.

Fig. 6. MLP Confusion Matrix



V. DISCUSSION

A. Technical Evolution: From Prototype to Hardened Auditor

This iteration shifts the project from simple data visualization toward Architectural Resiliency through a Defense-in-Depth strategy. By implementing a SHA-256 Cryptographic Kill Switch, the system ensures data provenance; if an adversary attempts to poison the ingestion layer, the system immediately severs the link. This reflects a "binary trust" model where data is either authenticated or discarded as a liability.

Additionally, the transition to Robust Scaling is a strategic defensive choice. Unlike standard normalisation, which is easily skewed by extreme values, Robust Scaling utilises the median and interquartile range. This prevents attackers from "blinding" the AI with malicious outliers, ensuring that the 0.83 ROC-AUC reliability is maintained even in contested cyber-physical environments.

B. Weaponisation Scenarios and Tactical Sabotage

The strategic necessity of these improvements is best illustrated through two primary weaponisation paths:

- 1) The Exhausted Pilot Scenario (Data Poisoning): Attackers manipulate biometric logs to mask physiological exhaustion. Without the SHA-256 integrity check and Z-score filtering, command dashboards would falsely report a pilot as "Rested," risking the catastrophic deployment of a compromised operator in control of multi-million dollar national assets.
- 2) The Secret Base Scenario (Pattern of Life): Even harmless step counts can inadvertently reveal clandestine installations. FitnessGuard utilises Salted Hashing and Differential Privacy principles to mathematically obscure individual identities. This prevents "digital exhaust" from exposing mission-critical locations while still allowing for aggregate health insights.

VI. CONCLUSION: ETHICAL, SECURITY, AND TRUST GOVERNANCE

This study concludes that securing the human operator in a cyber-physical world is the defining challenge of the 2026 future. The transition to the Internet of Battlefield Things (IoBT) has created a "Privacy Paradox": the granular data required to optimise a soldier's readiness is the exact same data required for an adversary to end their life via precision targeting.

A. The "Black Box" Problem and Algorithmic Trust

A significant risk in autonomous warfare is the lack of transparency. When an AI flags a soldier as "High Risk," the commander must trust that the decision is based on genuine biometric compromise and not a "hallucination." Our results, featuring an ROC-AUC of 0.83 and a stable 0.54 log loss convergence, serve as a "Proof of Trust." They prove that the 32-16-8 MLP has moved beyond random guessing to a stable metabolic understanding.

B. Societal Ethics and the Right to Anonymity

Beyond the battlefield, the general population faces the threat of "Pattern of Life" (PoL) surveillance. FitnessGuard demonstrates that privacy and utility can coexist. By implementing Salted Hashing, we prove that a system can deliver life-saving health interventions without creating a "surveillance state." Ultimately, as the lines between physical and digital reality disappear, our security must move beyond defending physical borders to defending the integrity of the human data itself.

VII. MITIGATIONS AND STRATEGIC RECOMMENDATIONS

To transition toward a "Trustworthy IoT" infrastructure, we propose the following technical mitigations and policy recommendations based on the FitnessGuard blueprint:

1) Mandate Cryptographic Provenance (Zero-Trust Ingestion)

Mitigation: We implemented a SHA-256 Integrity Gate to resolve the issue of "silent" data corruption.

Recommendation: National security agencies and healthcare providers must mandate that every biometric packet be signed with a digital seal at the sensor level. Our solution proves that a "Kill Switch" (st.stop) is the only logical response to a hash mismatch, preventing poisoned data from ever influencing a command decision.

2) Standardise Adversarial Statistical Filtering

Mitigation: FitnessGuard utilises Z-score anomaly detection via SciPy to act as a mathematical "bouncer."

Recommendation: AI auditors must be required to filter out data that is physically impossible (e.g., $\$z > 3\$$). This protects the neural network from "Data Poisoning" attempts designed to skew the AI's metabolic baseline.

3) Implement Deep-Layer Hardening for High-Stakes AI

Mitigation: We evolved the model from a shallow prototype to a 32-16-8 MLP architecture.

Recommendation: For high-stakes national security applications, "shallow" models are a liability. We recommend a minimum three-layer bottleneck structure to provide the computational depth required to detect the subtle, non-linear inconsistencies of an adversary attempting to mask operator exhaustion.

4) Promote Algorithmic Fairness through Synthetic Injection

Mitigation: To mitigate bias, we used Synthetic Bio-Data Injection via NumPy to balance the dataset across ages and genders.

Recommendation: To protect the general population, governments must mandate that AI models are audited for demographic fairness. This ensures that age or gender do not become "hidden vulnerabilities" that a hostile actor can exploit to time their tactical escalations.

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